Radxa ZERO 3W Schematic

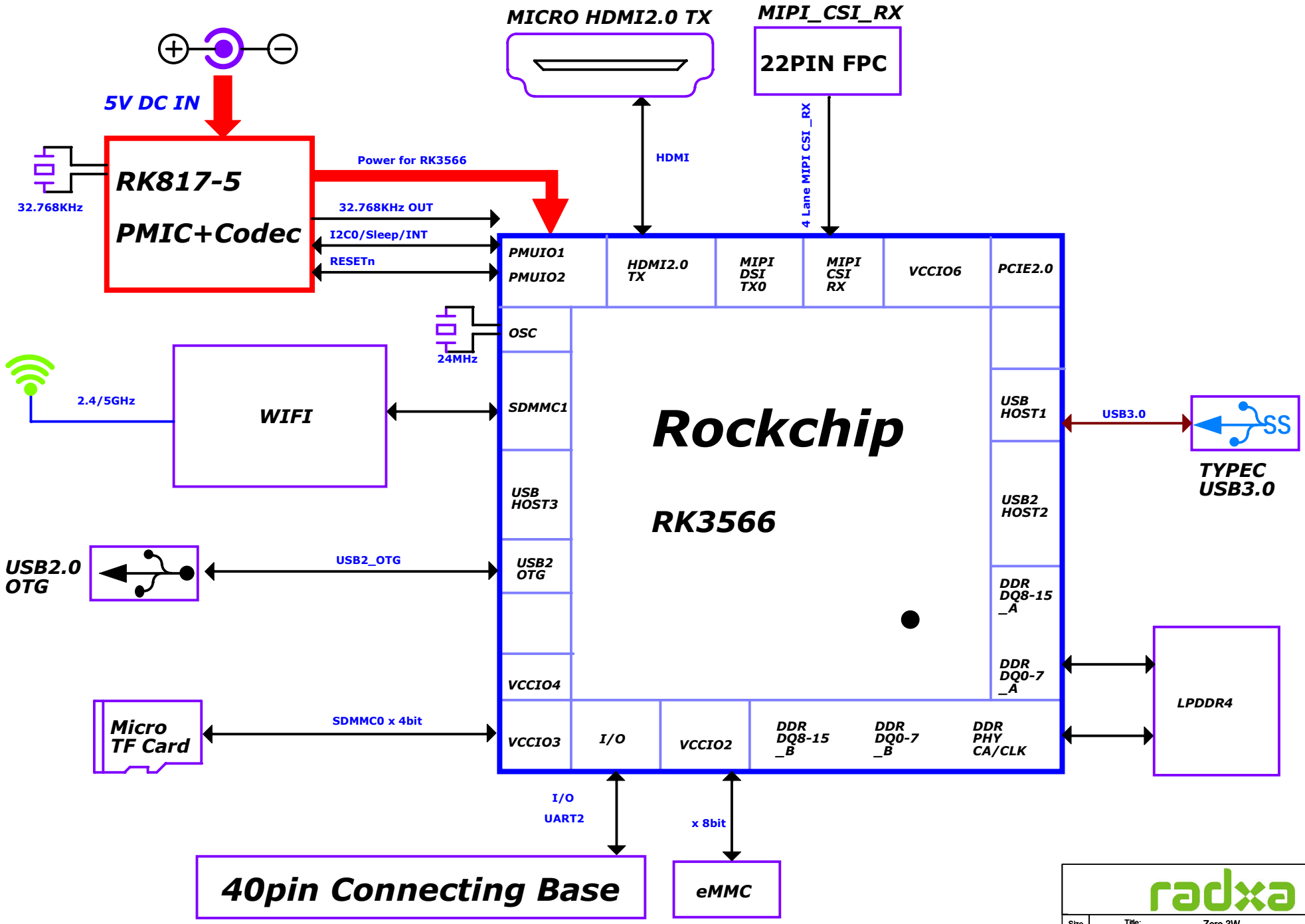
V1.12

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22.Power_DC IN

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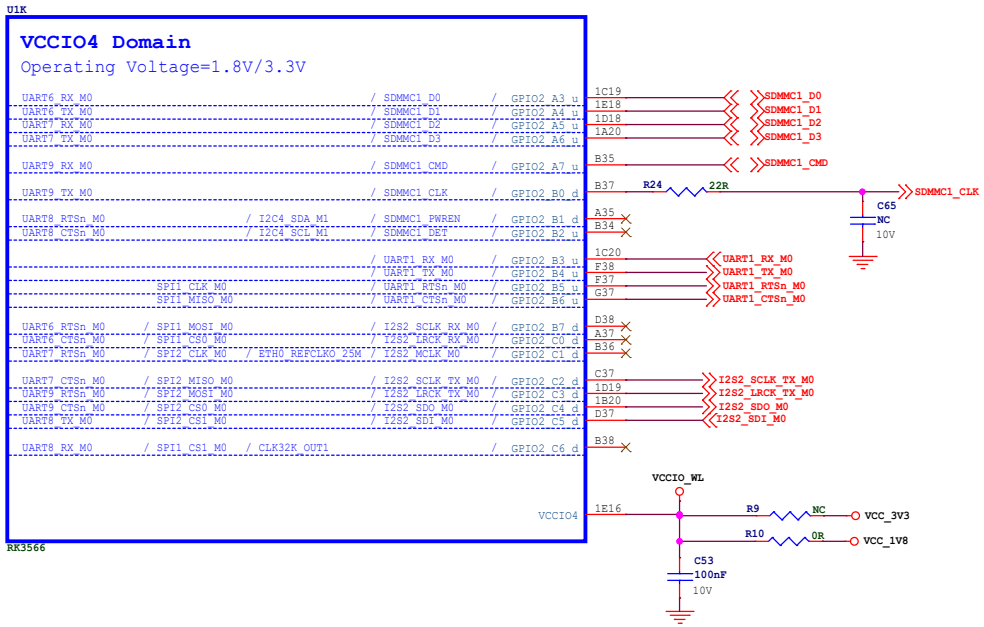
ROCK 3 Zero W Ref Block Diagram



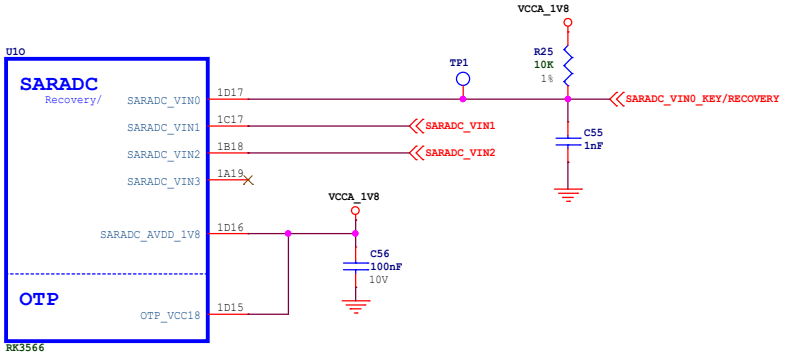
radxa

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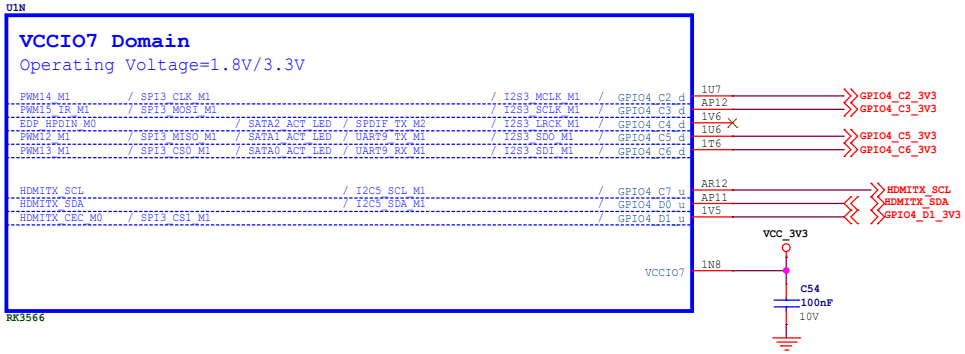
RK3566_K(VCCIO4 Domain)

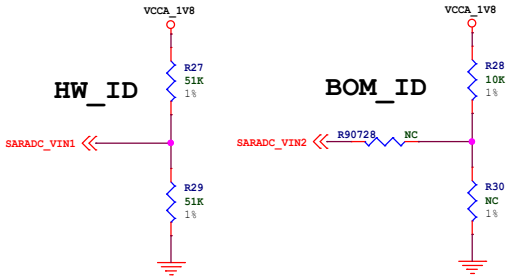


RK3566_O(SARADC/OTP)



RK3566_N(VCCIO7 Domain)





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RK3566_U

U1U

USB2.0 OTG_0
HS/FS/LS

(Download Port)

USB_OTG0_DP — R37 — USB_OTG0_DP
USB_OTG0_DM — R38 — USB_OTG0_DM
USB_OTG0_VBUSDET — T38 — USB_OTG0_VBUSDET
USB_OTG0_ID — T37 — USB_OTG0_ID

C44 100nF 10V

USB2.0 HOST_1
HS/FS/LS

(DP/DM for USB3.0 HOST)

USB_HOST1_DP — 1J19 — USB3_HOST1_DP
USB_HOST1_DM — 1J20 — USB3_HOST1_DM

USB2.0 Power
(for OTG_0 and HOST_1)

USB_AVDD1_0V9 — 1J17 — USB3_AVDD_0V9 — R19 1R 1% — VDDA_0V9
USB_AVDD1_1V8 — 1H16 — USB3_AVDD_1V8 — R20 1R 1% — VCCA_1V8
USB_AVDD1_3V3 — 1H18 — USB3_AVDD_3V3 — R21 0R — VCC_3V3

C45 100nF 10V
C46 100nF 10V
C47 100nF 10V

MULTI PHY U_S
(SS for USB3.0 HOST or SATA3_1)

USB3_HOST1_SSTXp/SATA1_TXp — W37 — USB3_HOST1_SSTXP
USB3_HOST1_SSTXn/SATA1_TXn — W38 — USB3_HOST1_SSTXN
USB3_HOST1_SSRXp/SATA1_RXp — V38 — USB3_HOST1_SSRXP
USB3_HOST1_SSRXn/SATA1_RXn — V37 — USB3_HOST1_SSRXN

MULTI PHY P_S
(PCIE2.0 or SATA3_2)

PCIE20_TXp/SATA2_TXp — AA37
PCIE20_TXn/SATA2_TXn — AA38
PCIE20_RXp/SATA2_RXp — AB37
PCIE20_RXn/SATA2_RXn — AB38
PCIE20_REFCLKp — 1K19
PCIE20_REFCLKn — 1K20

MULTI PHY Power
(for MULTI PHY U_S and P_S)

MULTI_PHY_AVDD_0V9 — 1K16 — VDDA_0V9
MULTI_PHY_AVDD_1V8 — 1K17 — VCCA_1V8

C51 100nF 10V
C52 100nF 10V

RK3566

RK3566_V

U1V

USB2.0
(for HOST_2 and HOST_3)

USB_HOST2_DP — V2 — USB2_HOST2_DP
USB_HOST2_DM — V1 — USB2_HOST2_DM
USB_HOST3_DP — Y2
USB_HOST3_DM — Y1

USB_AVDD2_0V9 — 1J4 — USB3_AVDD_0V9
USB_AVDD2_1V8 — 1J3 — USB3_AVDD_1V8
USB_AVDD2_3V3 — 1J2 — USB3_AVDD_3V3

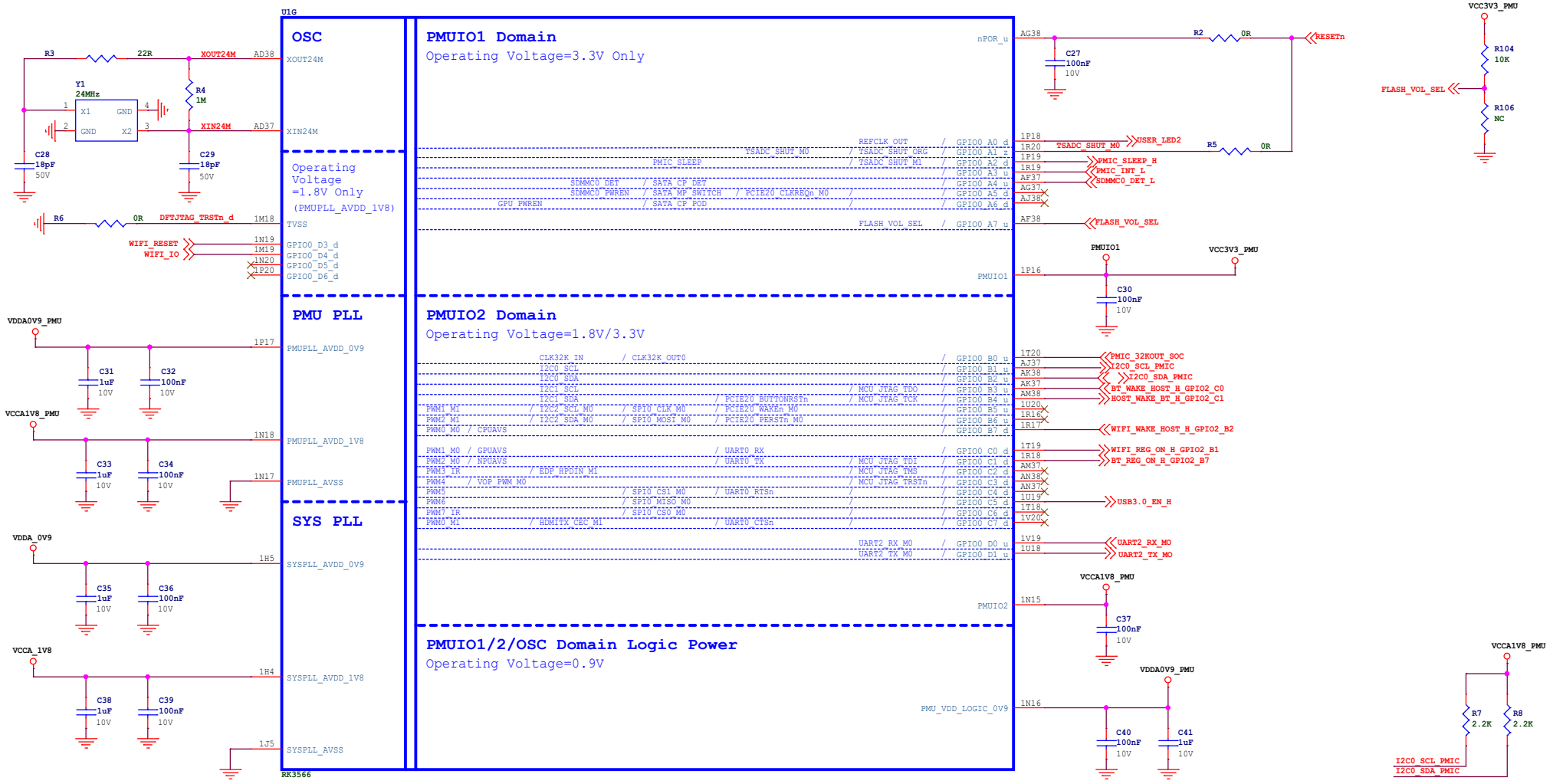
C48 DNP 10V
C49 DNP 10V
C50 DNP 10V

RK3566

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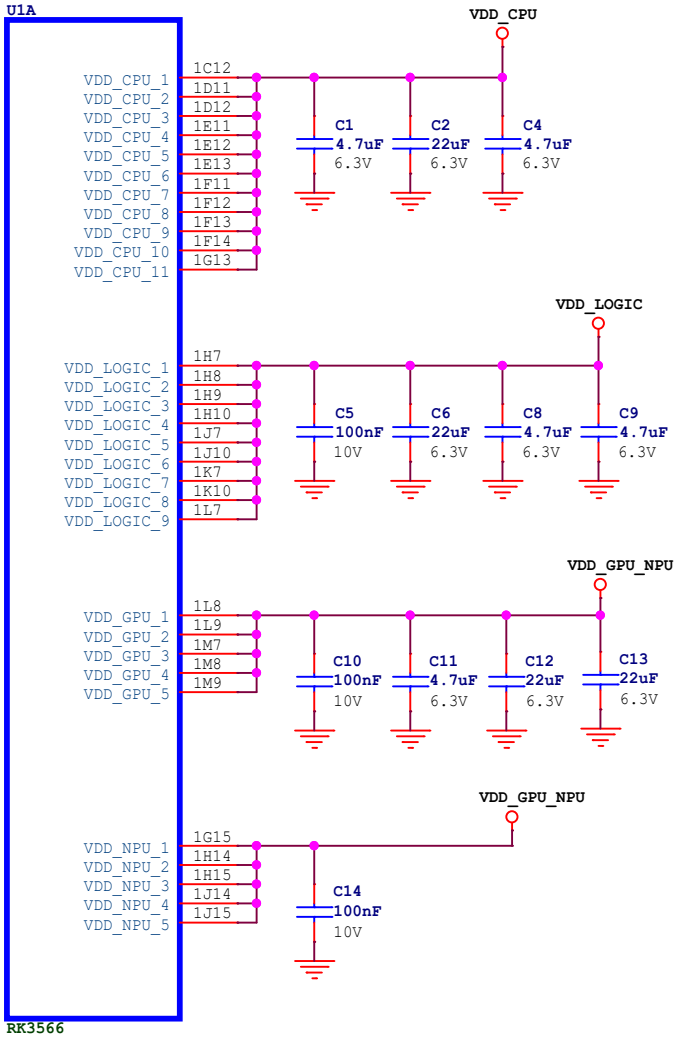
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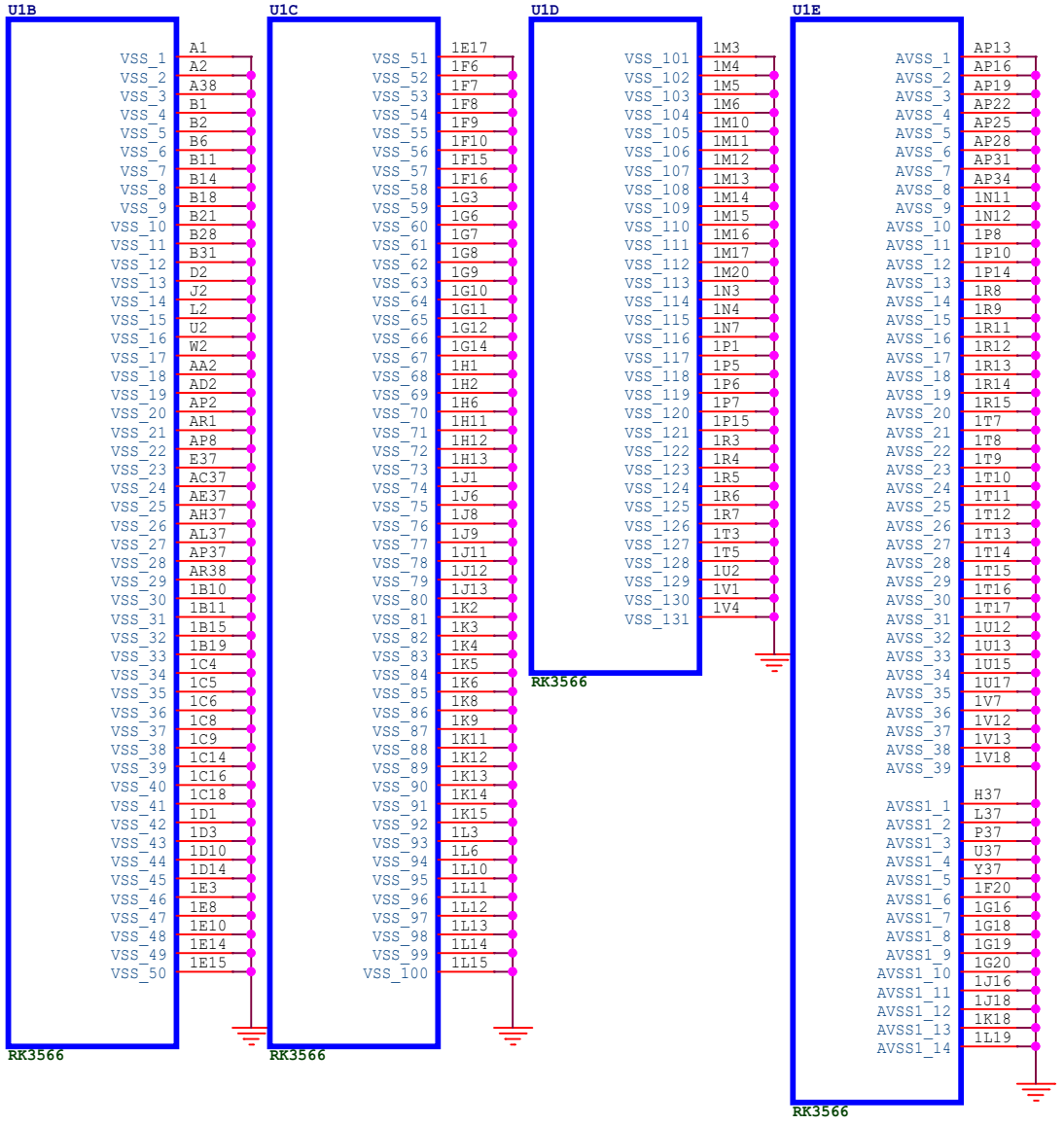
RK3566(OSC/PLL/PMUIO1/2)



Size: A3	Title: Zero 3W	REV: V1.12
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RK3566_ABCDE
(Power&GND)

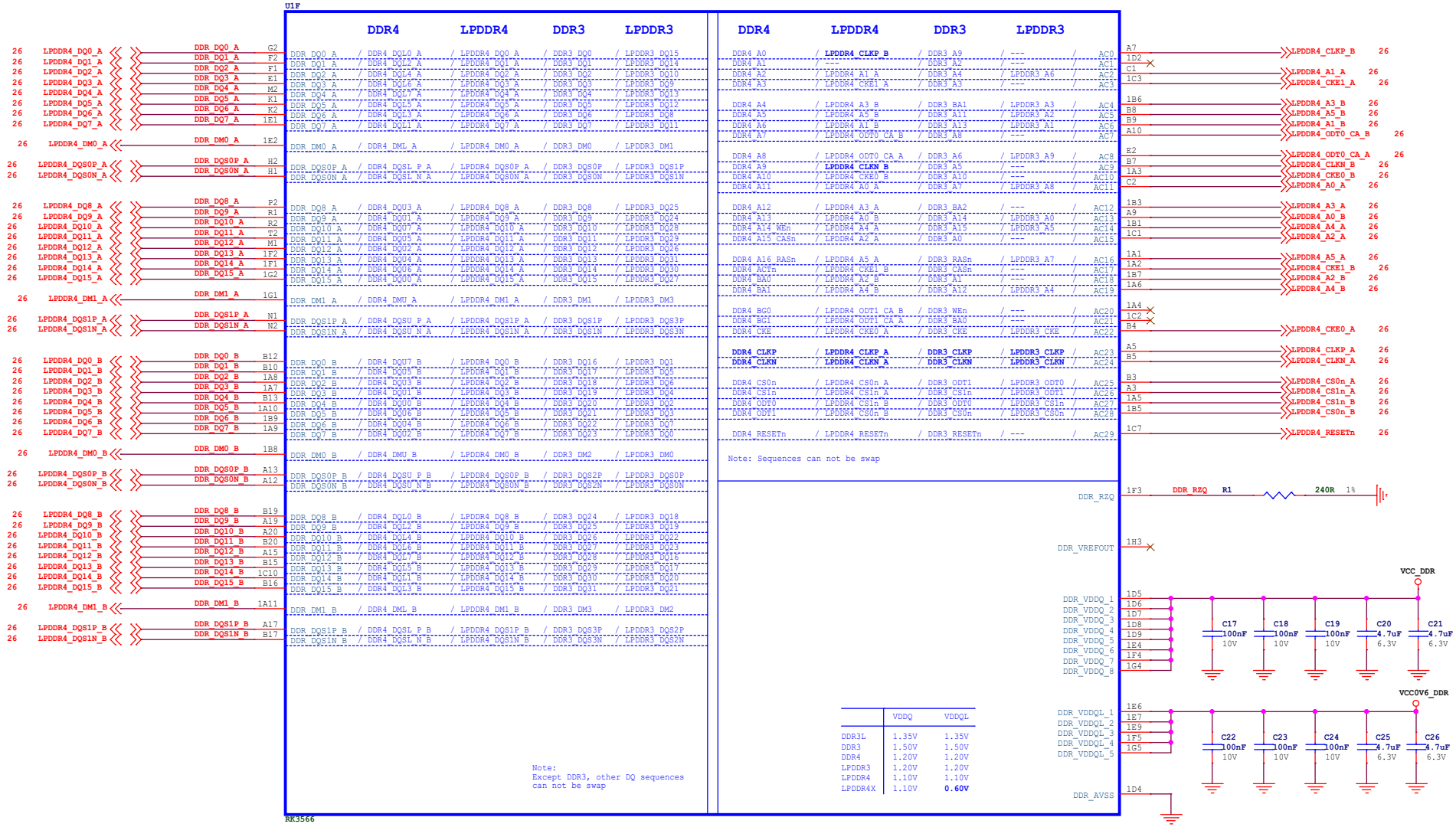




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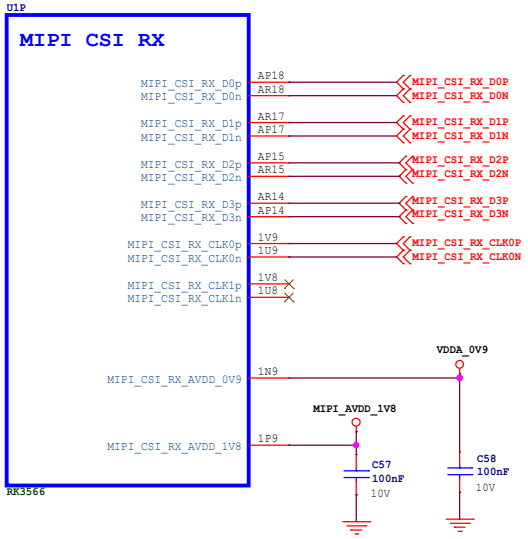
Size	Title:	Zero 3W	REV
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RK3566_F(DDR PHY)

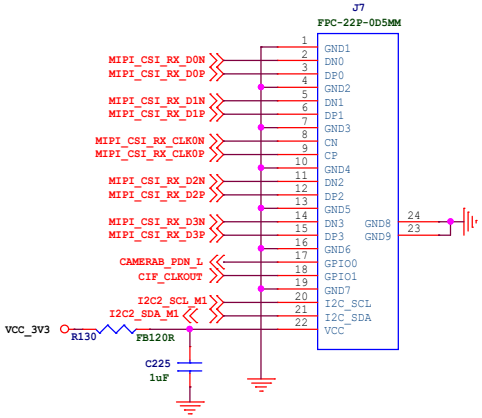


Size	Title:	Zero 3W	REV
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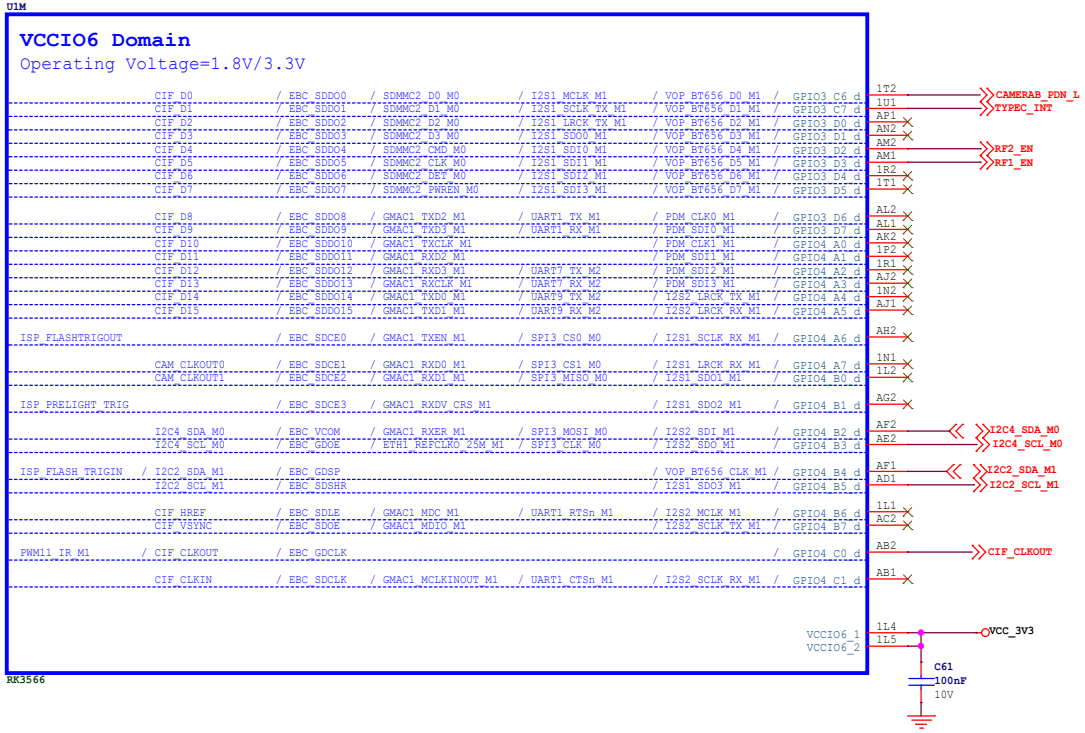
RK3566_P(MIPI_CSI_RX)

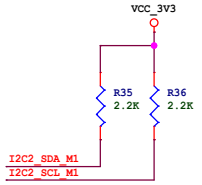


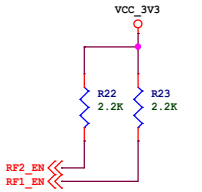
MIPI_CSI_RX 4Lanes

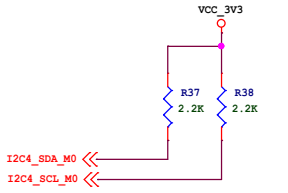


RK3566_M(VCCIO6 Domain)



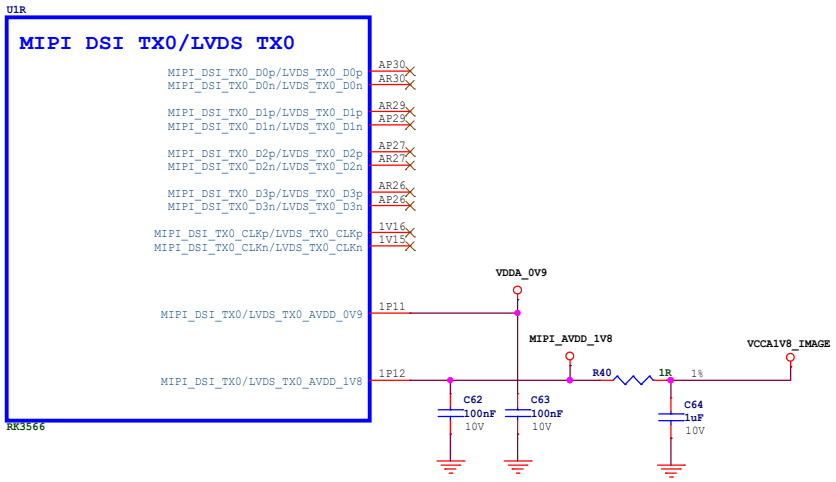




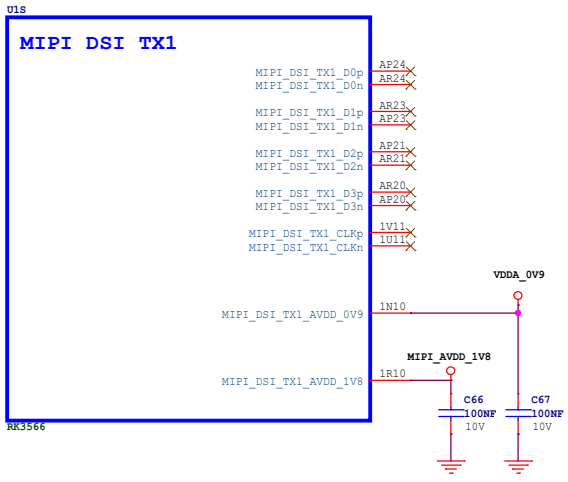


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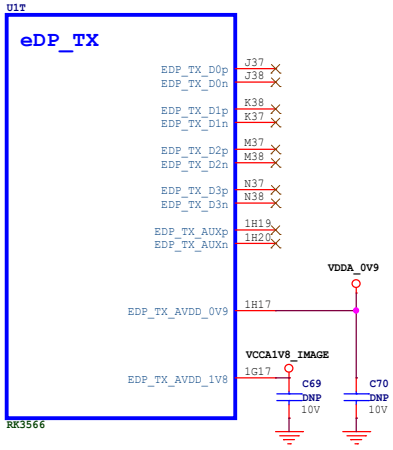
RK3566_R(MIPI_DSI_TX0/LVDS_TX0)



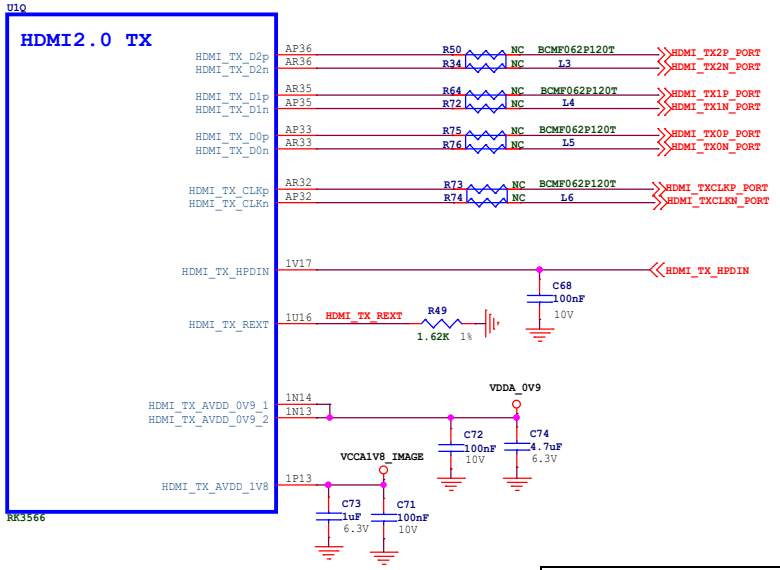
RK3566_S(MIPI_DSI_TX1/LVDS_TX1)



RK3566_T(eDP/DP TX)



RK3566_Q(HDMI2.0 TX)



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RK3566_L(VCCIO5 Domain)

U1L

VCCIO5 Domain

Operating Voltage=1.8V/3.3V

	VOP_BT1120_D0	/ SPI1_CS0_M1			/ SDMMC2_D0_M1	/ GPIO3_A1_d
	VOP_BT1120_D1		/ GMAC1_TXD2_M0	/ I2S3_MCLK_M0	/ SDMMC2_D1_M1	/ GPIO3_A2_d
	VOP_BT1120_D2		/ GMAC1_TXD3_M0	/ I2S3_SCLK_M0	/ SDMMC2_D2_M1	/ GPIO3_A3_d
	VOP_BT1120_D3		/ GMAC1_RXD2_M0	/ I2S3_LRCK_M0	/ SDMMC2_D3_M1	/ GPIO3_A4_d
	VOP_BT1120_D4		/ GMAC1_RXD3_M0	/ I2S3_SDO_M0	/ SDMMC2_CMD_M1	/ GPIO3_A5_d
	VOP_BT1120_CLK		/ GMAC1_TXCLK_M0	/ I2S3_SDI_M0	/ SDMMC2_CLK_M1	/ GPIO3_A6_d
	VOP_BT1120_D5		/ GMAC1_RXCLK_M0		/ SDMMC2_DET_M1	/ GPIO3_A7_d
	VOP_BT1120_D6		/ ETH1_REFCLKO_25M_M0		/ SDMMC2_PWREN_M1	/ GPIO3_B0_d
PWM8_M0	/ VOP_BT1120_D7		/ GMAC1_RXD0_M0	/ UART4_RX_M1		/ GPIO3_B1_d
PWM9_M0	/ VOP_BT1120_D8		/ GMAC1_RXD1_M0	/ UART4_TX_M1		/ GPIO3_B2_d
	VOP_BT1120_D9	/ I2C5_SCL_M0	/ GMAC1_RXDV_CRS_M0		/ PDM_SDI0_M2	/ GPIO3_B3_d
	VOP_BT1120_D10	/ I2C5_SDA_M0	/ GMAC1_RXER_M0		/ PDM_SDI1_M2	/ GPIO3_B4_d
PWM10_M0	/ VOP_BT1120_D11	/ I2C3_SCL_M1	/ GMAC1_TXD0_M0			/ GPIO3_B5_d
PWM11_IR_M0	/ VOP_BT1120_D12	/ I2C3_SDA_M1	/ GMAC1_TXD1_M0			/ GPIO3_B6_d
PWM12_M0			/ GMAC1_TXEN_M0	/ UART3_TX_M1	/ PDM_SDI2_M2	/ GPIO3_B7_d
PWM13_M0			/ GMAC1_MCLKINOUT_M0	/ UART3_RX_M1	/ PDM_SDI3_M2	/ GPIO3_C0_d
	VOP_BT1120_D13	/ SPI1_MOSI_M1		/ PCIE20_PERSTn_M1	/ I2S1_SDO2_M2	/ GPIO3_C1_d
	VOP_BT1120_D14	/ SPI1_MISO_M1		/ UART5_TX_M1	/ I2S1_SDO3_M2	/ GPIO3_C2_d
	VOP_BT1120_D15	/ SPI1_CLK_M1		/ UART5_RX_M1	/ I2S1_SCLK_RX_M2	/ GPIO3_C3_d
PWM14_M0		/ VOP_PWM_M1	/ GMAC1_MDC_M0	/ UART7_TX_M1	/ PDM_CLK1_M2	/ GPIO3_C4_d
PWM15_IR_M0		/ SPDIF_TX_M1	/ GMAC1_MDIO_M0	/ UART7_RX_M1	/ I2S1_LRCK_RX_M2	/ GPIO3_C5_d

RK3566

1U5 AR10 AP10 AR9 AP9 1U4 1V3 AR7 AP7 AR6 AP6 1U3 1T4 1V2 AP5 AR4 AP4 AP3 AR2 1P3 1P4

GPIO3_A1_3V3 GPIO3_A2_3V3 GPIO3_A3_3V3 GPIO3_A4_3V3 GPIO3_A5_3V3 GPIO3_A6_3V3 GPIO3_A7_3V3 GPIO3_B0_3V3 GPIO3_B1_3V3 GPIO3_B2_3V3 GPIO3_B3_3V3 GPIO3_B4_3V3 I2C3_SCL_M1 I2C3_SDA_M1 GMAC1_INT/PMEB_GPIO3_B7 GMAC1_RSTn_GPIO3_C0 GPIO3_C1_3V3 GPIO3_C2_3V3 GPIO3_C3_3V3 GPIO3_C4_3V3

VCC_3V3

VCCIO5_1 1N5
VCCIO5_2 1N6

C75
100nF
10V

VCC_3V3

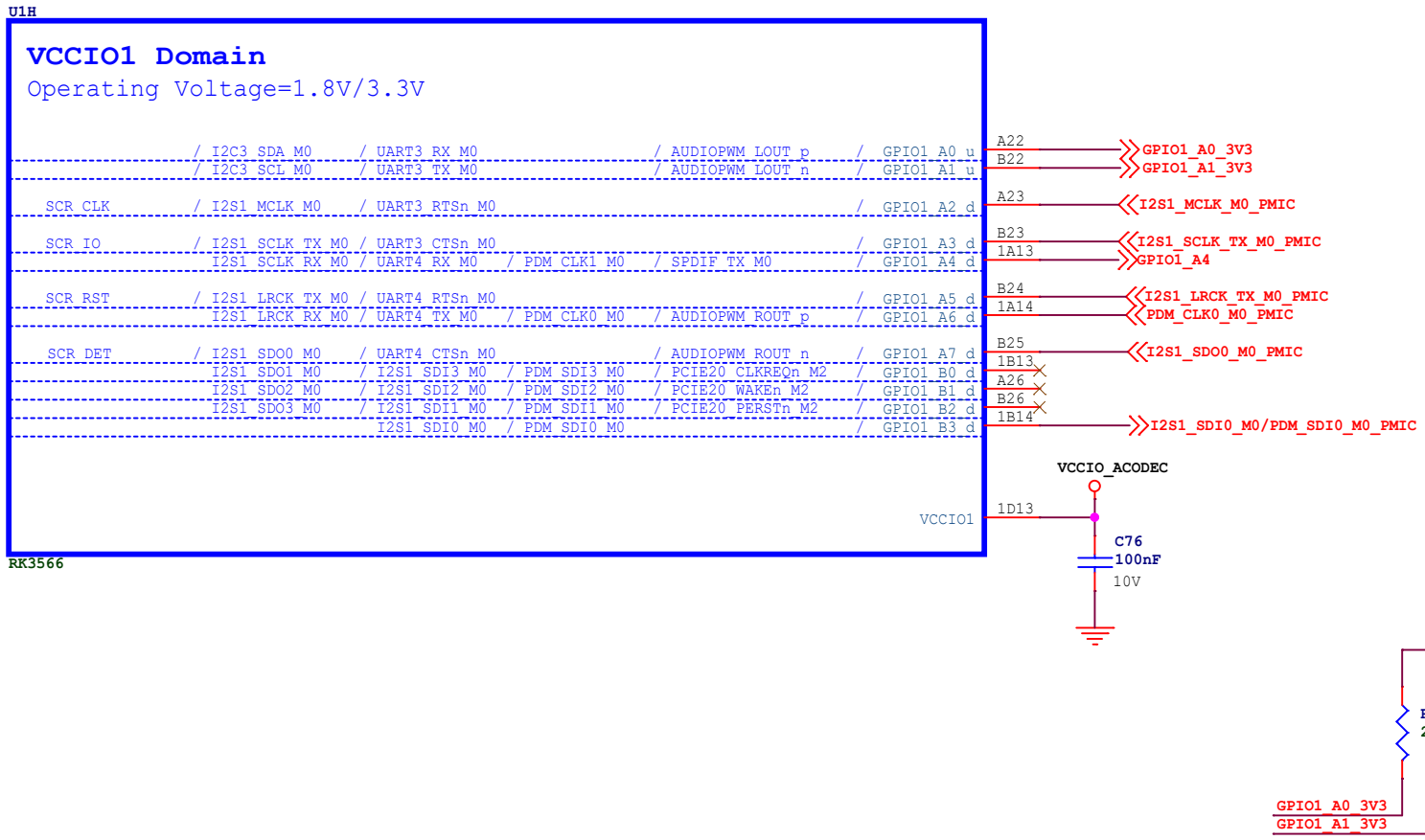
R52 2.2K
R53 2.2K

I2C3_SCL_M1
I2C3_SDA_M1

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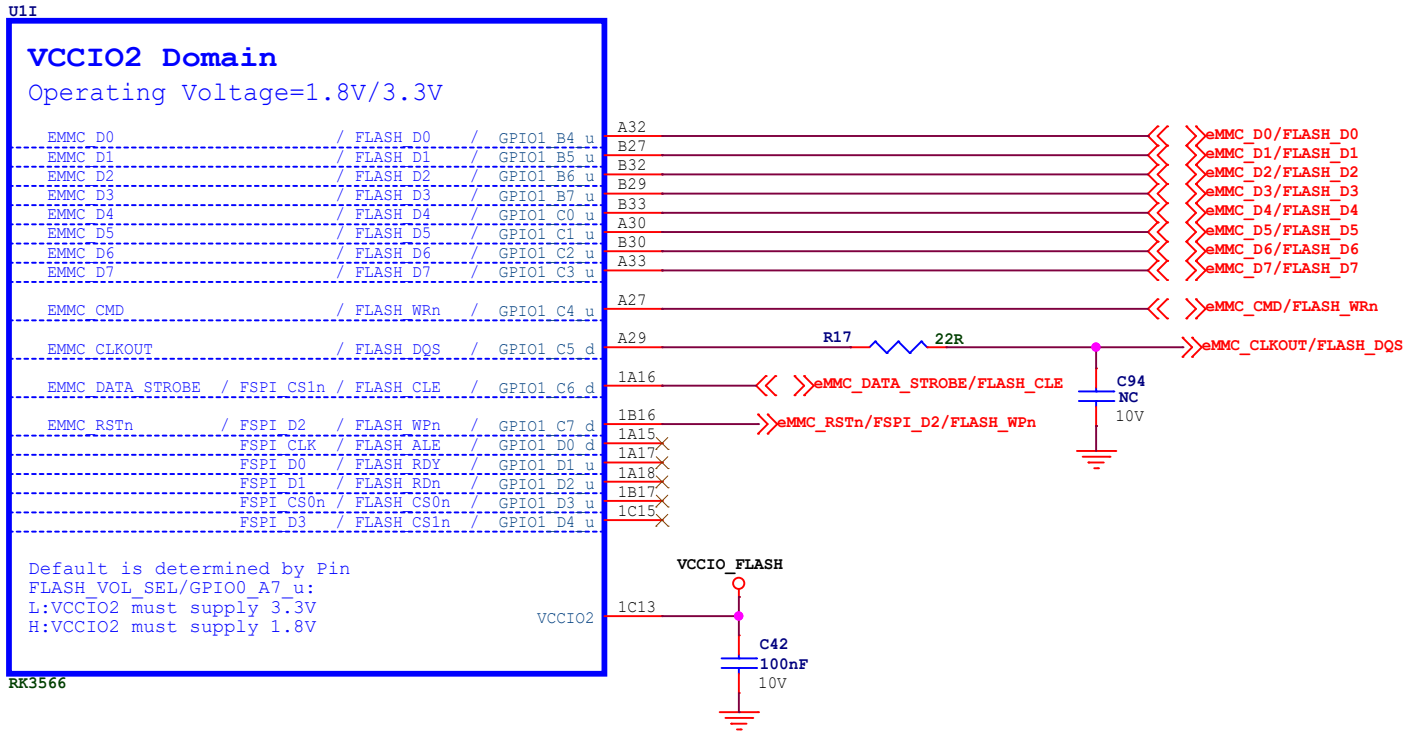
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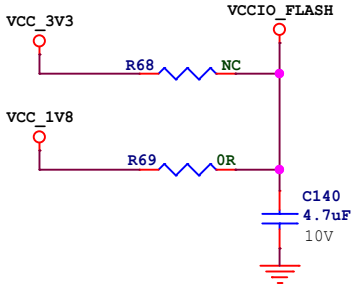
RK3566_H(VCCIO1 Domain)



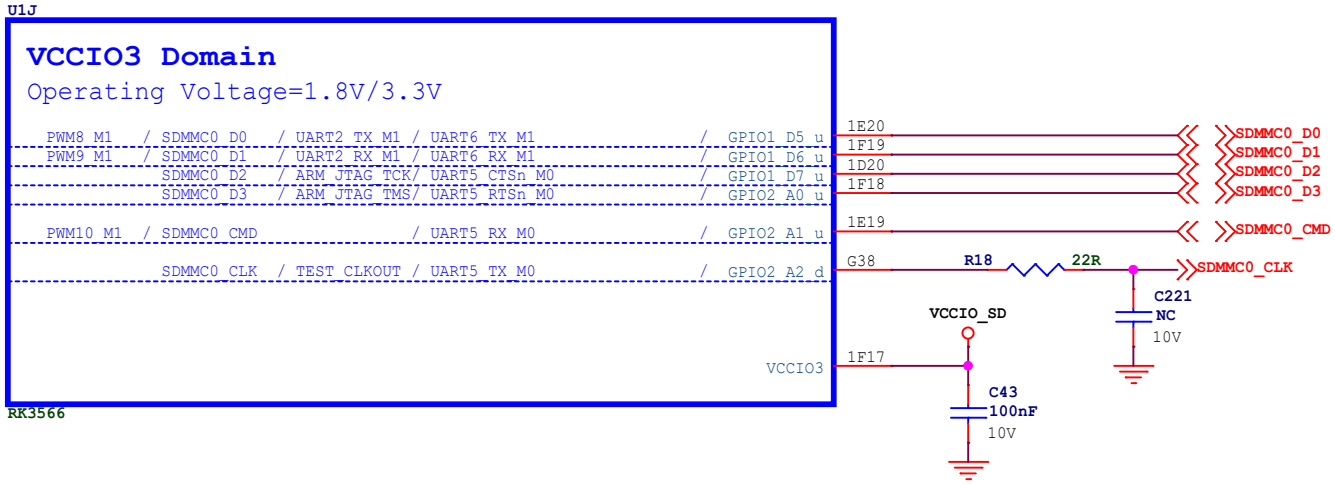
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RK3566_I(VCCIO2 Domain)

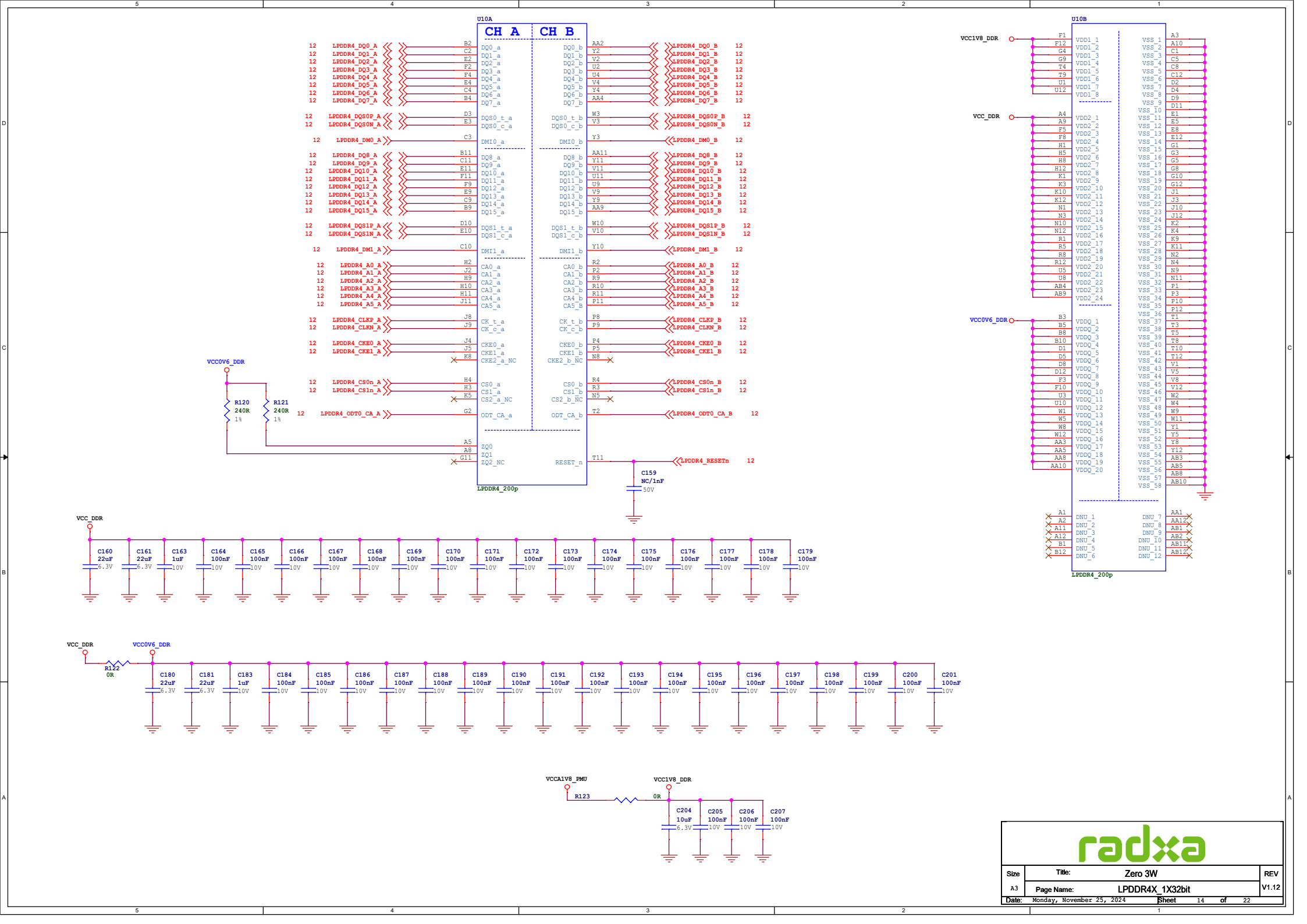




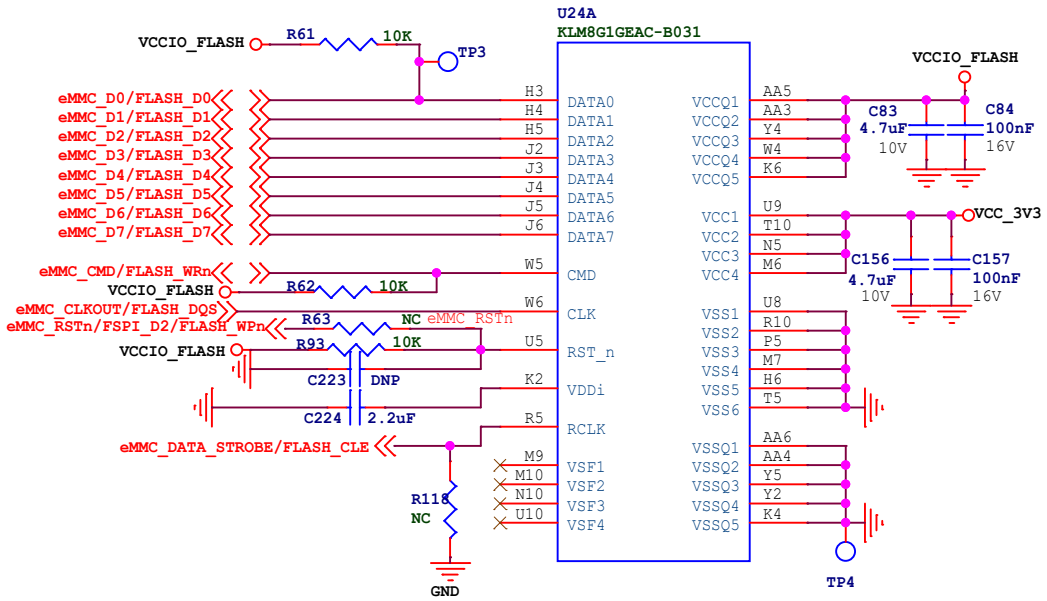
RK3566_J(VCCIO3 Domain)

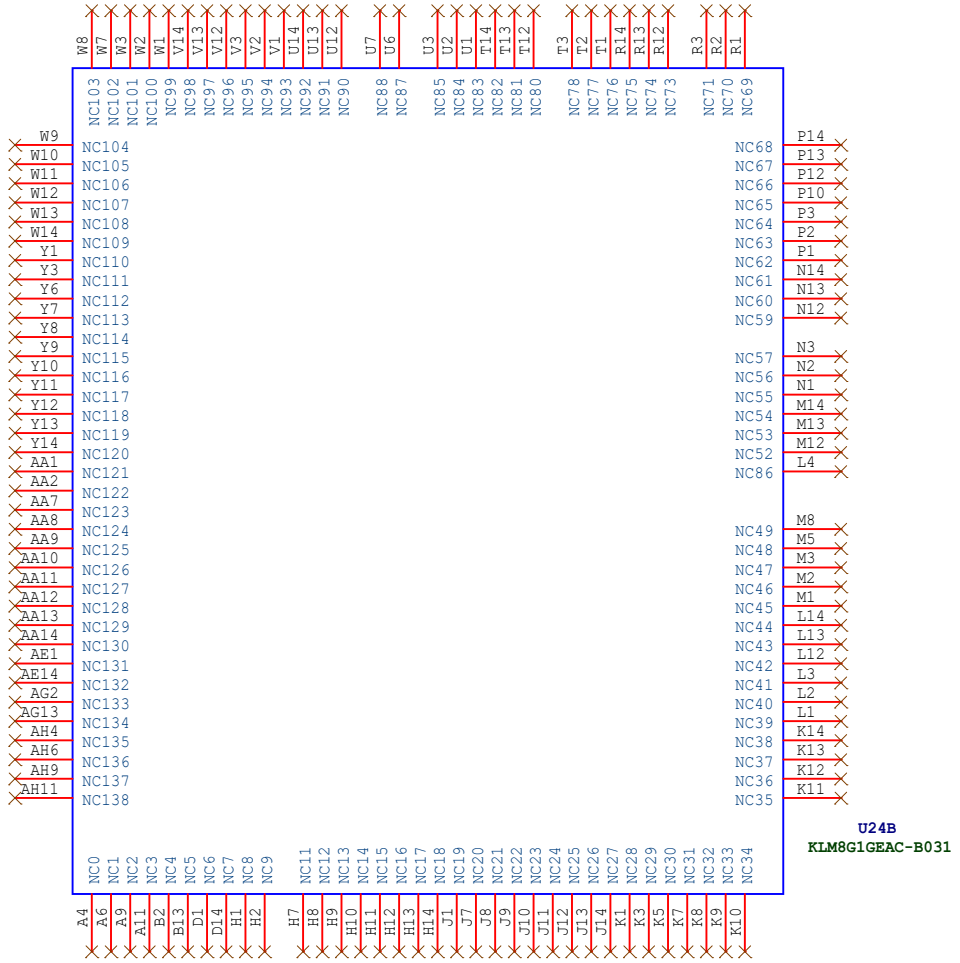


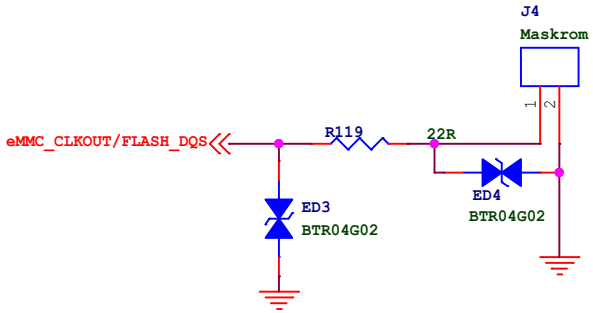
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U10A
CH A
CH B
LPDDR4_200p
U10B
LPDDR4_200p
VCC1V8_DDR
VCC_DDR
VCC0V6_DDR
VCCA1V8_PMU
R120 240R 1%
R121 240R 1%
R122 0R
R123 0R
C159 NC/1nF 50V
LPDDR4_DQ0_A
LPDDR4_DQ1_A
LPDDR4_DQ2_A
LPDDR4_DQ3_A
LPDDR4_DQ4_A
LPDDR4_DQ5_A
LPDDR4_DQ6_A
LPDDR4_DQ7_A
LPDDR4_DQS0P_A
LPDDR4_DQS0N_A
LPDDR4_DM0_A
LPDDR4_DQ8_A
LPDDR4_DQ9_A
LPDDR4_DQ10_A
LPDDR4_DQ11_A
LPDDR4_DQ12_A
LPDDR4_DQ13_A
LPDDR4_DQ14_A
LPDDR4_DQ15_A
LPDDR4_DQS1P_A
LPDDR4_DQS1N_A
LPDDR4_DM1_A
LPDDR4_A0_A
LPDDR4_A1_A
LPDDR4_A2_A
LPDDR4_A3_A
LPDDR4_A4_A
LPDDR4_A5_A
LPDDR4_CLKP_A
LPDDR4_CLKN_A
LPDDR4_CKE0_A
LPDDR4_CKE1_A
LPDDR4_CS0n_A
LPDDR4_CS1n_A
LPDDR4_ODT0_CA_A
LPDDR4_RESETn
LPDDR4_DQ0_B
LPDDR4_DQ1_B
LPDDR4_DQ2_B
LPDDR4_DQ3_B
LPDDR4_DQ4_B
LPDDR4_DQ5_B
LPDDR4_DQ6_B
LPDDR4_DQ7_B
LPDDR4_DQS0P_B
LPDDR4_DQS0N_B
LPDDR4_DM0_B
LPDDR4_DQ8_B
LPDDR4_DQ9_B
LPDDR4_DQ10_B
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LPDDR4_DQ15_B
LPDDR4_DQS1P_B
LPDDR4_DQS1N_B
LPDDR4_DM1_B
LPDDR4_A0_B
LPDDR4_A1_B
LPDDR4_A2_B
LPDDR4_A3_B
LPDDR4_A4_B
LPDDR4_A5_B
LPDDR4_CLKP_B
LPDDR4_CLKN_B
LPDDR4_CKE0_B
LPDDR4_CKE1_B
LPDDR4_CS0n_B
LPDDR4_CS1n_B
LPDDR4_ODT0_CA_B
C160 22uF 6.3V
C161 22uF 6.3V
C163 1uF 10V
C164 100nF 10V
C165 100nF 10V
C166 100nF 10V
C167 100nF 10V
C168 100nF 10V
C169 100nF 10V
C170 100nF 10V
C171 100nF 10V
C172 100nF 10V
C173 100nF 10V
C174 100nF 10V
C175 100nF 10V
C176 100nF 10V
C177 100nF 10V
C178 100nF 10V
C179 100nF 10V
C180 22uF 6.3V
C181 22uF 6.3V
C183 1uF 10V
C184 100nF 10V
C185 100nF 10V
C186 100nF 10V
C187 100nF 10V
C188 100nF 10V
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C190 100nF 10V
C191 100nF 10V
C192 100nF 10V
C193 100nF 10V
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C195 100nF 10V
C196 100nF 10V
C197 100nF 10V
C198 100nF 10V
C199 100nF 10V
C200 100nF 10V
C201 100nF 10V
C204 10uF 6.3V
C205 100nF 10V
C206 100nF 10V
C207 100nF 10V
radxa
Size A3
Title: Zero 3W
REV V1.12
Page Name: LPDDR4X_1X32bit
Date: Monday, November 25, 2024
Sheet 14 of 22

EMMC FLASH

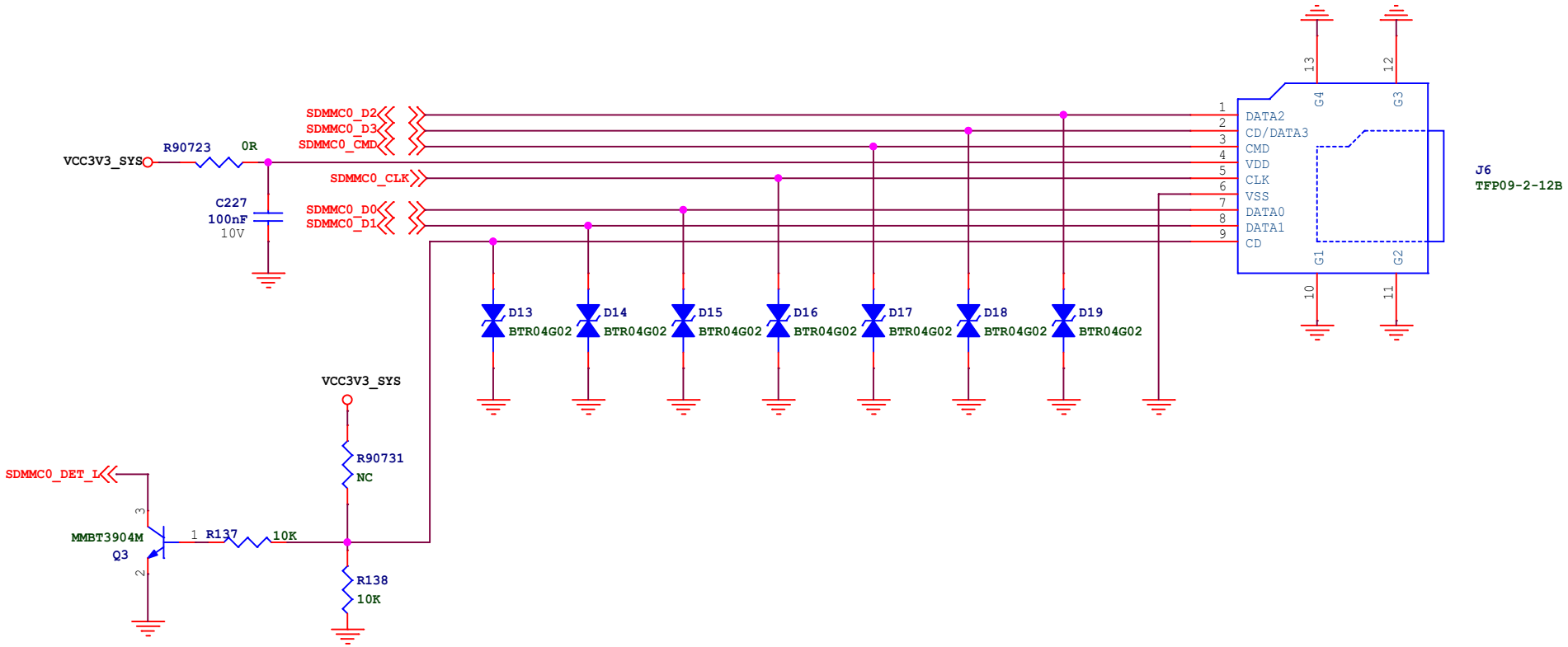




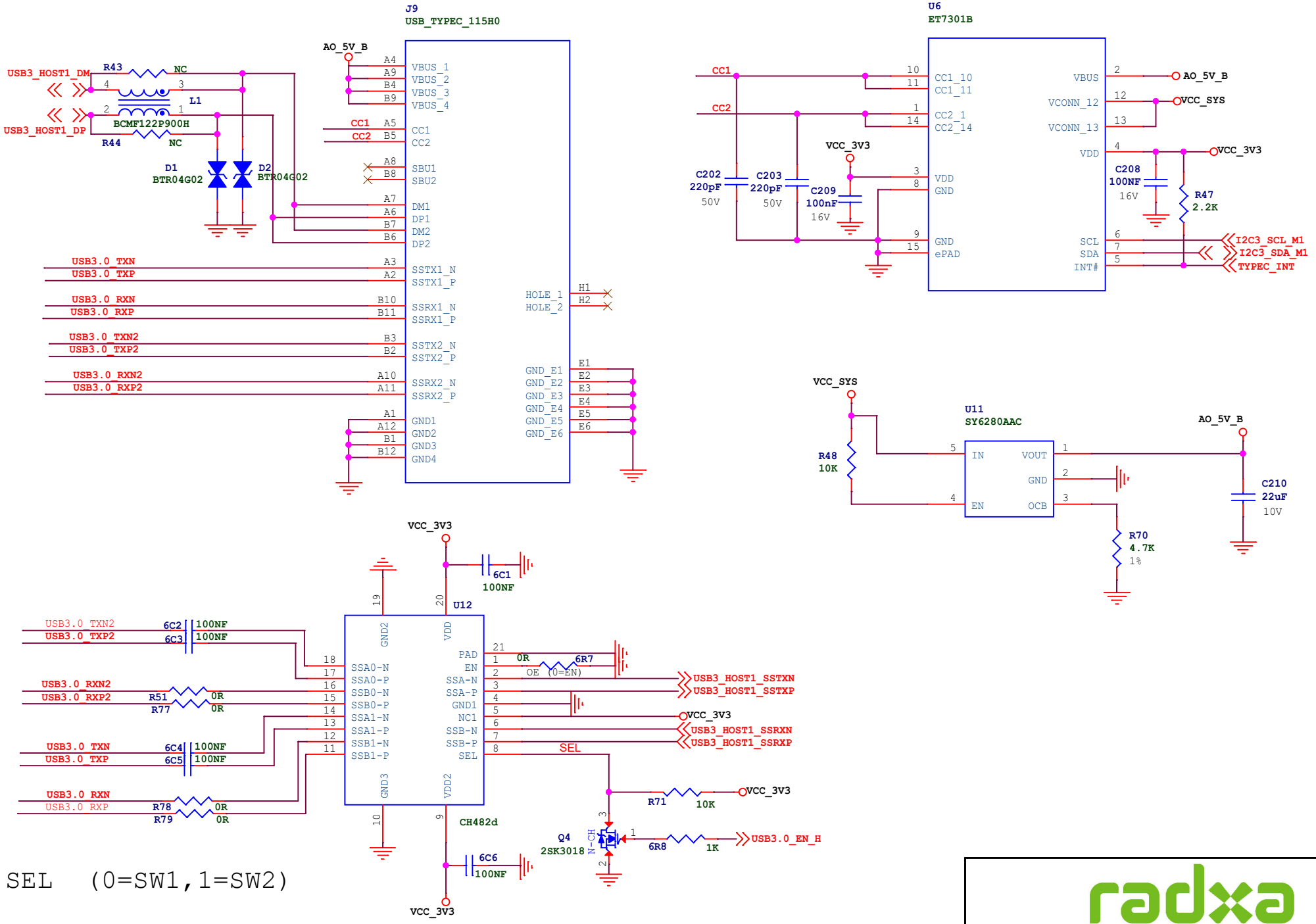


Size	Title:	Zero 3W	REV
A4	Page Name:	SPI Flash/eMMC	V1.12
Date:	Monday, November 25, 2024	Sheet 15 of 22	

MicroSD Card

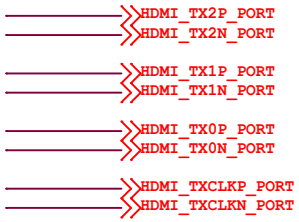


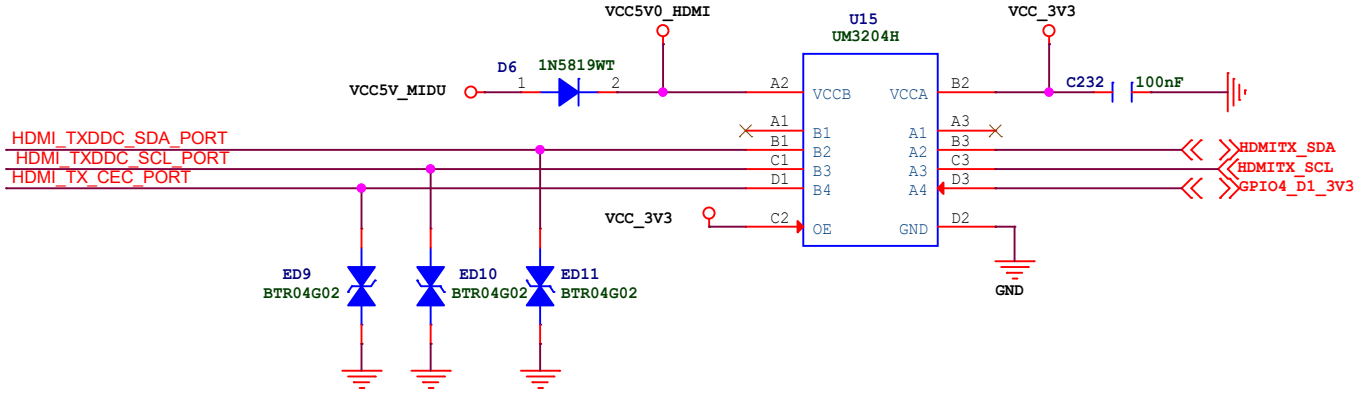
Size	Title:	Zero 3W	REV
A4	Page Name:	MicroSD Card	V1.12
Date:	Monday, November 25, 2024	Sheet 16 of 22	



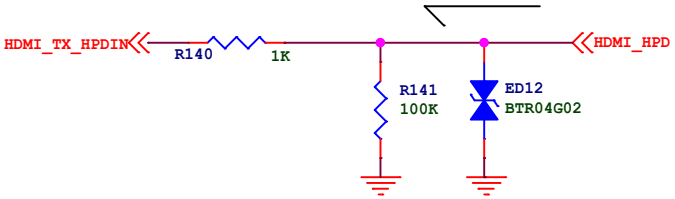
SEL (0=SW1,1=SW2)

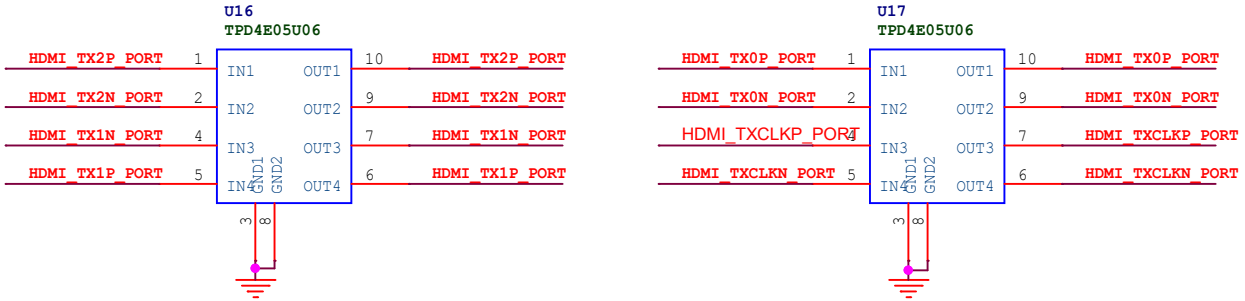
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A4	Page Name:	USB Port	V1.12
Date:	Monday, November 25, 2024	Sheet 17 of 22	



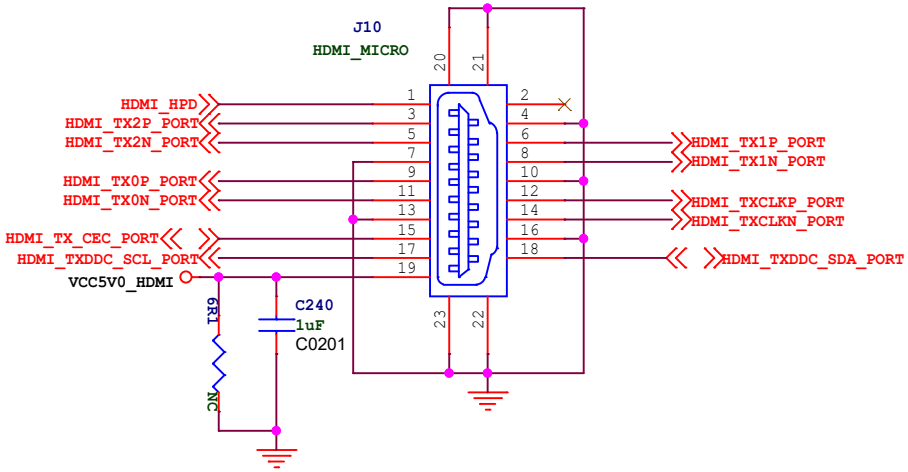


HDMI TX HPD





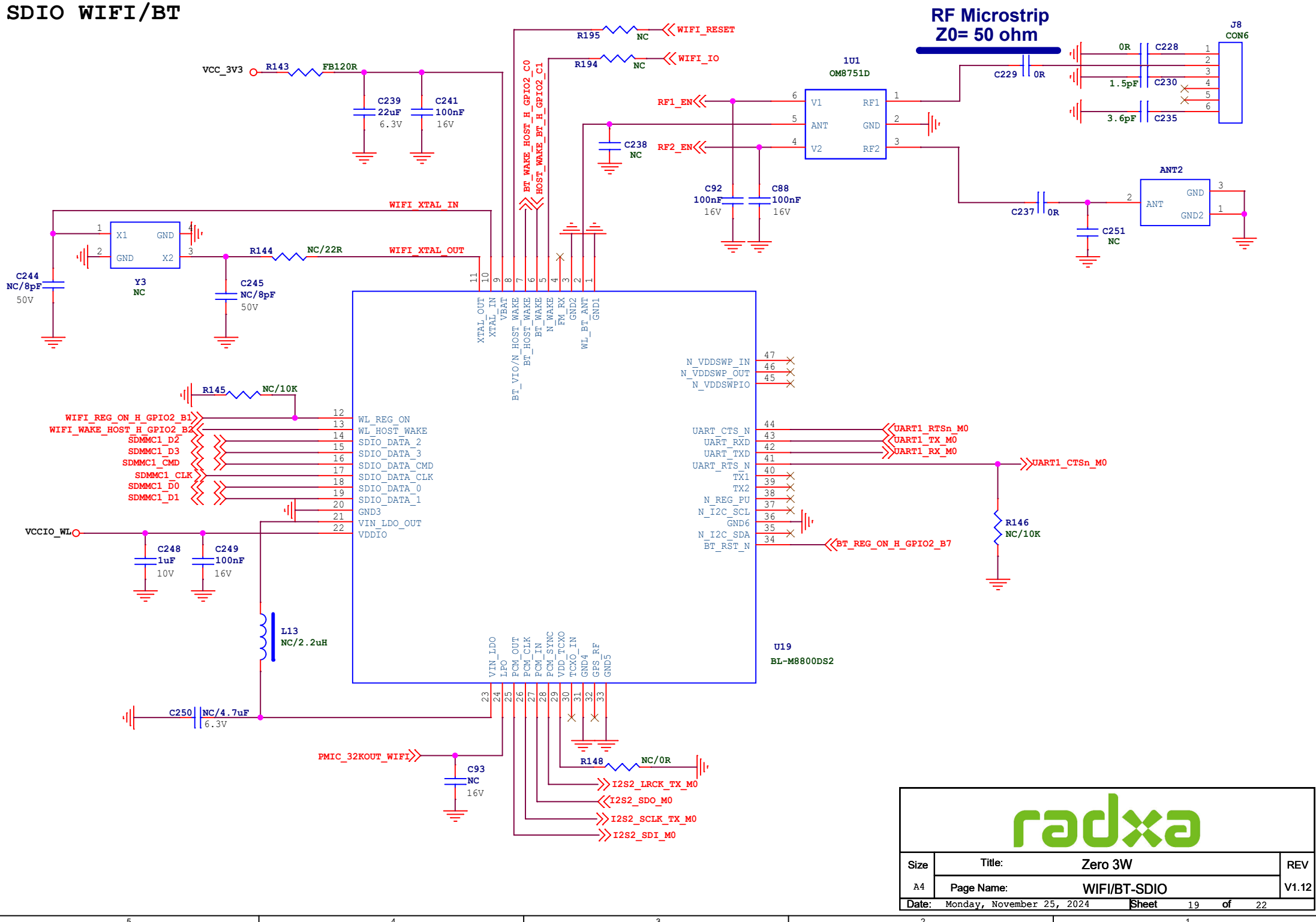
MICRO HDMI output



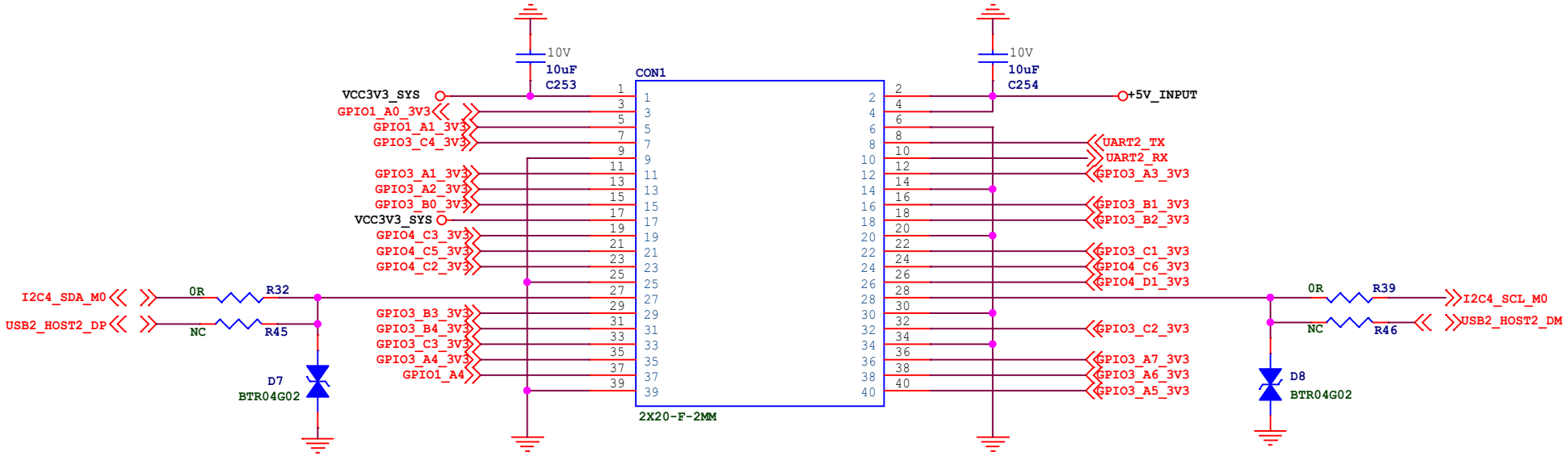
radxa

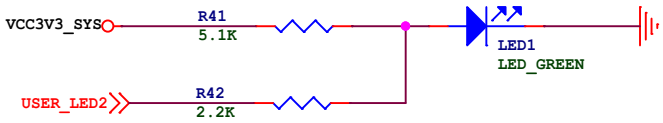
Size	Title:	Zero 3W	REV
A4	Page Name:	HDMI2.0 TX	V1.12
Date:	Monday, November 25, 2024	Sheet 18 of 22	

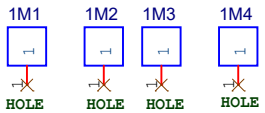
SDIO WIFI/BT

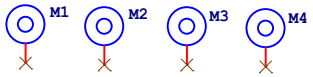


Size	Title:	Zero 3W	REV
A4	Page Name:	WIFI/BT-SDIO	V1.12
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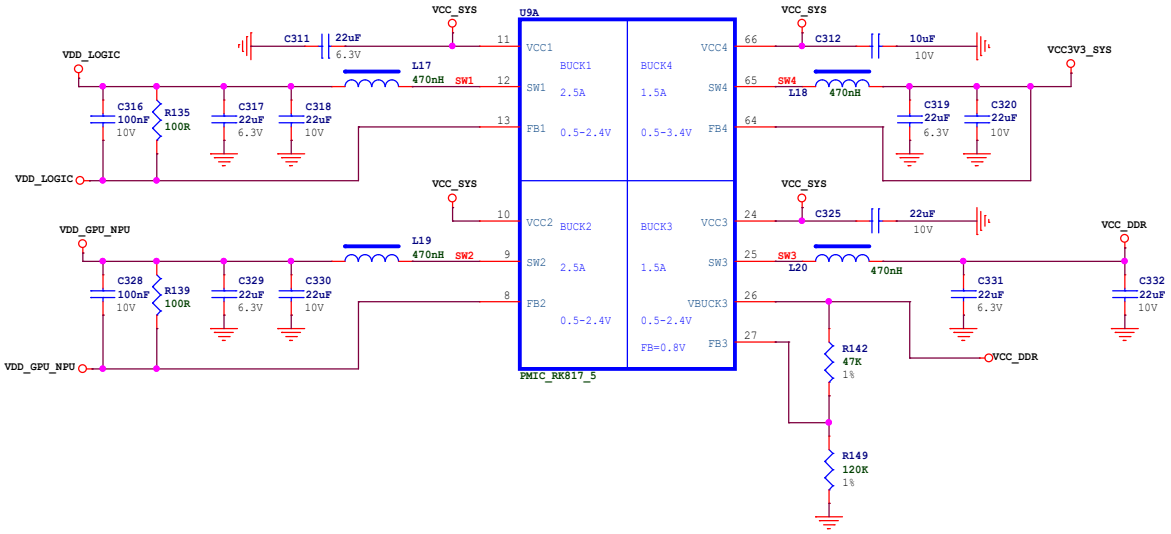




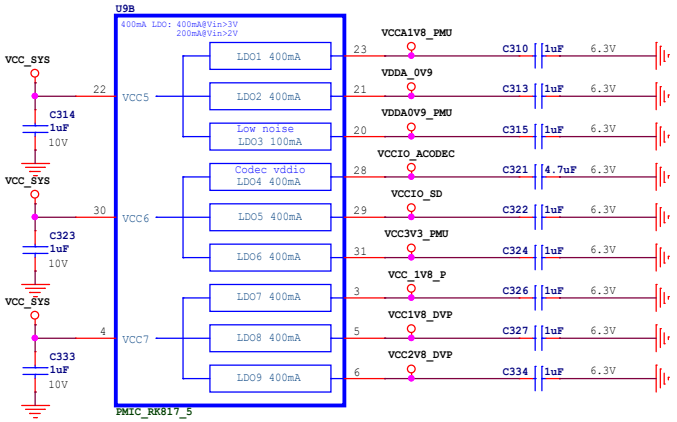


Size	Title:	Zero 3W	REV
A4	Page Name:	CONNECTOR	V1.12
Date:	Monday, November 25, 2024	Sheet 20 of 22	

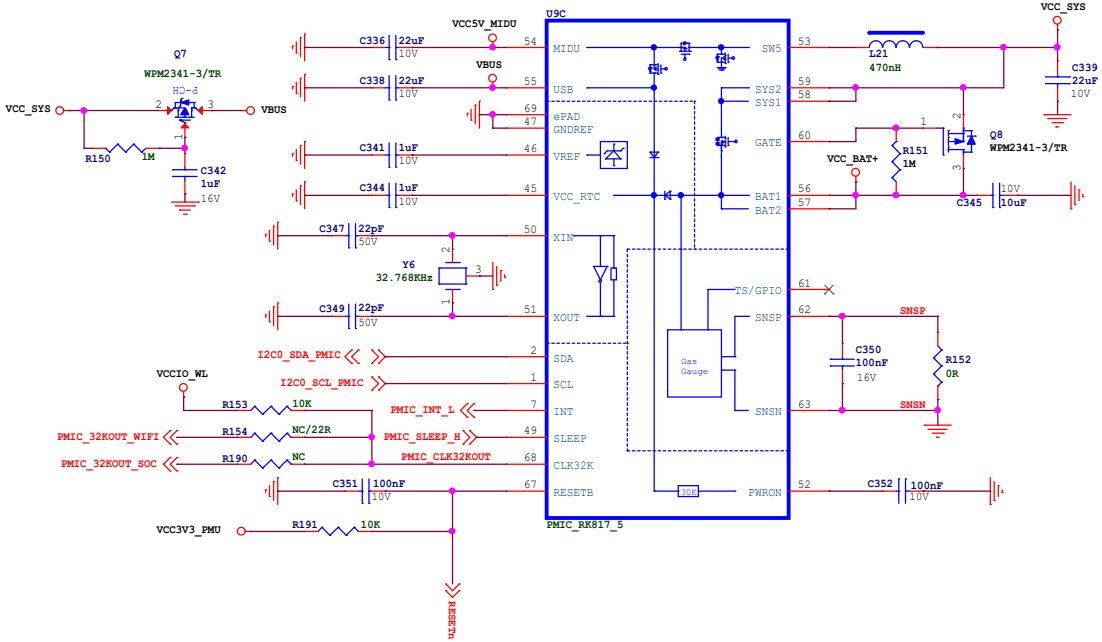
PMIC RK817 DCDC



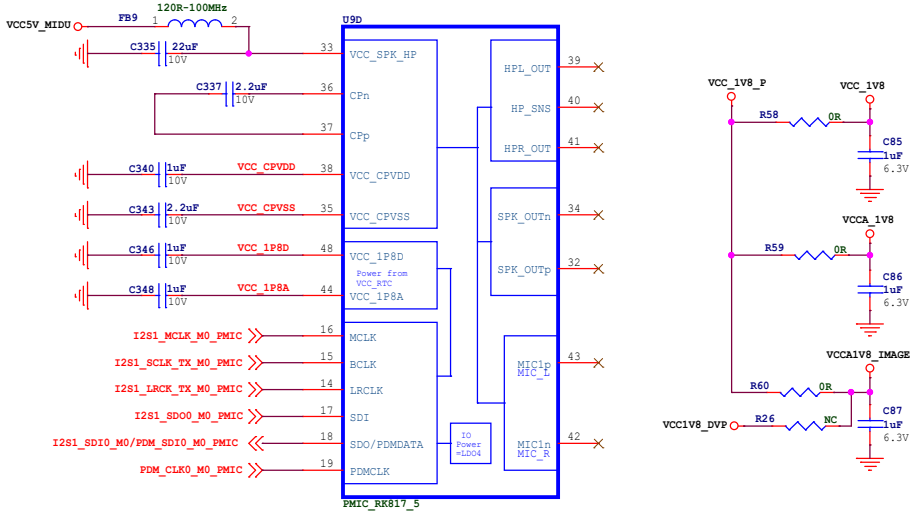
PMIC RK817 LDO



PMIC RK817 Managerment



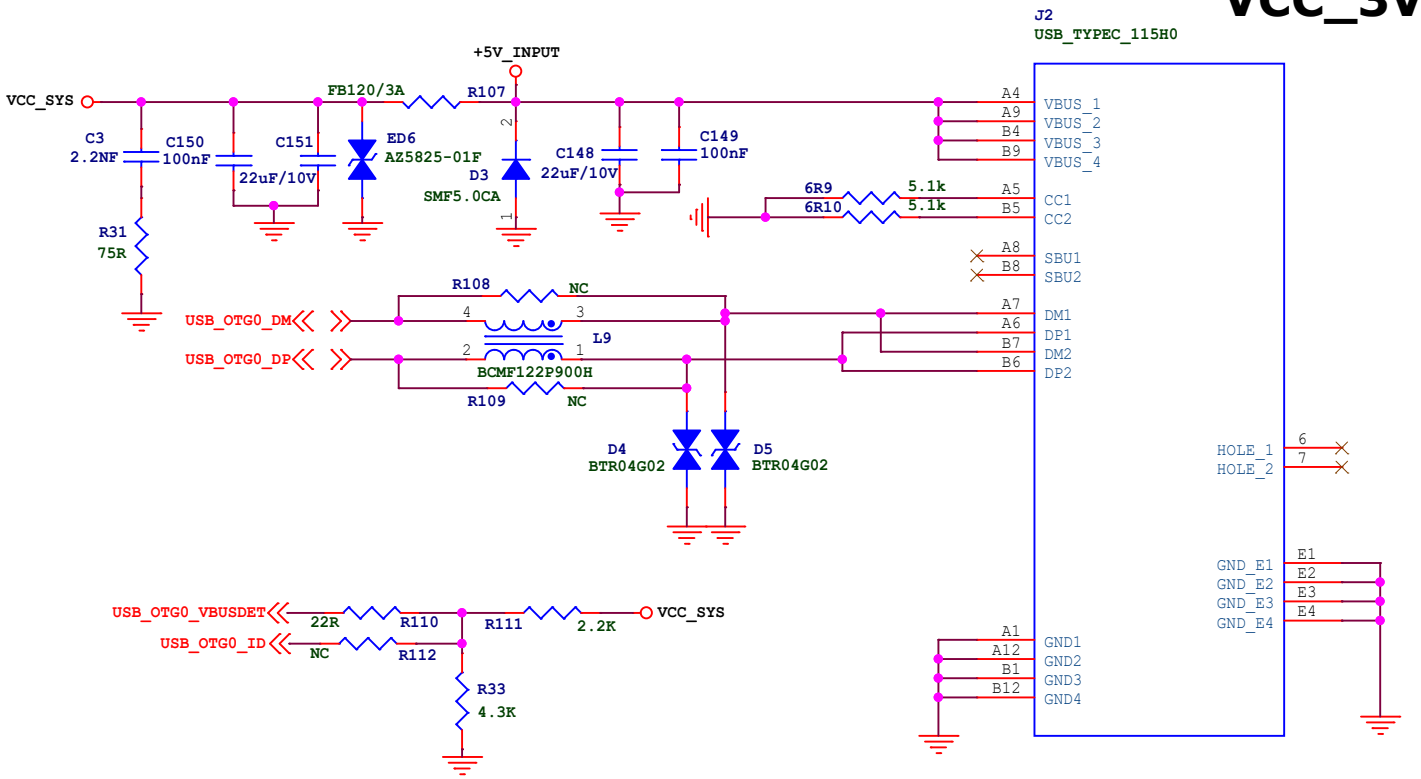
PMIC RK817 CODEC



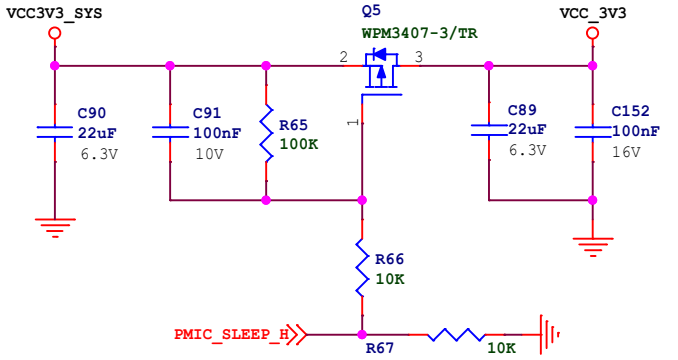
radxa

Size	Title:	REV
A3	Zero 3W	V1.12
	Page Name: Power_PMIC	
Date: Monday, November 25, 2024	Sheet 21 of 22	

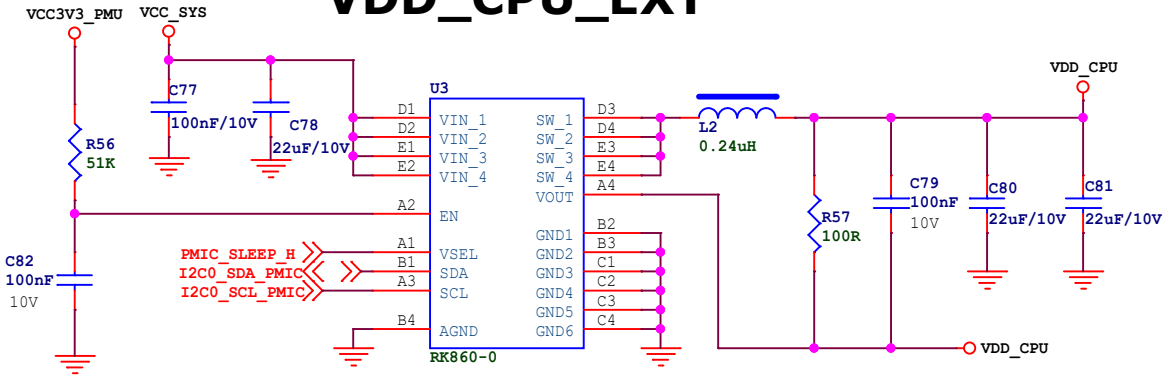
DC 5V IN



VCC_3V3 (Power OFF under SLEEP)



VDD_CPU_EXT



Size: A4	Title: Zero 3W	REV: V1.12
	Page Name: Power_DC IN	
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